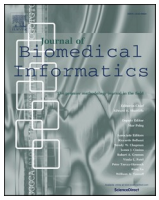
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Special Communication



Criteria2Query 3.0: Leveraging generative large language models for clinical trial eligibility query generation

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ABSTRACT

Objective: Automated identification of eligible patients is a bottleneck of clinical research. We propose Criteria2Query (C2Q) 3.0, a system that leverages GPT-4 for the semi-automatic transformation of clinical trial eligibility criteria text into executable clinical database queries.

Materials and Methods: C2Q 3.0 integrated three GPT-4 prompts for concept extraction, SQL query generation, and reasoning. Each prompt was designed and evaluated separately. The concept extraction prompt was benchmarked against manual annotations from 20 clinical trials by two evaluators, who later also measured SQL generation accuracy and identified errors in GPT-generated SQL queries from 5 clinical trials. The reasoning prompt was assessed by three evaluators on four metrics: readability, correctness, coherence, and usefulness, using corrected SQL queries and an open-ended feedback questionnaire.

Results: Out of 518 concepts from 20 clinical trials, GPT-4 achieved an F1-score of 0.891 in concept extraction. For SQL generation, 29 errors spanning seven categories were detected, with logic errors being the most common (n = 10; 34.48 %). Reasoning evaluations yielded a high coherence rating, with the mean score being 4.70 but relatively lower readability, with a mean of 3.95. Mean scores of correctness and usefulness were identified as 3.97 and 4.37, respectively.

Conclusion: GPT-4 significantly improves the accuracy of extracting clinical trial eligibility criteria concepts in C2Q 3.0. Continued research is warranted to ensure the reliability of large language models.

1. Introduction

The recent advances in generative large language models (LLMs) represented by ChatGPT and its subsequent model GPT-4 have shown remarkable performances in language understanding and generation [1,2]. A massive amount of textual data and model parameters enable the creation of comprehensive language models, resulting in human-like text generation [3–6]. These LLMs have demonstrated their potential in language generation, knowledge mining, and logical reasoning [7–10]. Hamer et al. evaluated the computability of the eligibility criteria using LLM for 4,135 criteria across 146 clinical trials and found that 72 % of the criteria were correctly identified using LLM [11]. Wong et al. evaluated GPT's ability to structure the eligibility criteria of 53 oncology trials and reported that GPT-4 achieved an F1-score from 64.8 to 65.4 on histology concept recognition and 66.7 to 72.5 on biomarker concept recognition [12]. Meanwhile, Peikos et al. leveraged ChatGPT to extract patient information from clinical notes and to match it with clinical trials [13]. Jin et al. developed TrialGPT, which employed GPT-3.5 to parse eligibility criteria and derive criterion-level relevance scores based on patient data [14]. TrialGPT was evaluated on three publicly available cohorts of 184 patients and 18,238 annotated clinical trials and outperformed the BERT-based models for ranking and excluding trials. These studies directly match patient records with clinical trials in GPT prompt. Despite impressive performance, as patient records in clinical databases accumulated over time, patient-trial matching systems need to be re-executed periodically, the proposed methods require substantial computing resources for using OpenAI API and must overcome formidable patient privacy protection challenges [15–17]. Therefore, in this study, we focus on using GPT to generate sharable cohort identification queries.

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Criteria2Query (C2Q) 1.0 is open-source software for automating information extraction and query formulation directly from free-text clinical trial eligibility criteria [18]. It combined machine-learning and rule-based methods to systematically parse criteria and convert them into SQL queries conforming to the Observational Medical Outcomes Partnership (OMOP)-Common Data Model (CDM) [19] C2Q 2.0 extended C2Q 1.0 by involving humans in the loop for more accurate concept recognition, entity and value normalization, and criteria simplification [20]. Despite the efforts previously made, the performance of both systems drops when parsing complex logical relationships among concepts [21–23]. In addition, the current method of C2Q 2.0 struggles to parse criteria comprising multiple measurements and respective upper and lower thresholds within a single sentence. Moreover, generally rule-based query formulation is hard to scale [24].

We hypothesize that comprehensive language understanding and generation abilities of LLM can overcome these challenges. In this paper, we present C2Q 3.0 incorporating OpenAI GPT-4, with a primary focus on enhancing information extraction and query formulation processes. We also intended to enhance system interpretability by providing users with a narrative system rationale. Herein, our methodological contributions include 1) comparing GPT's parsing performance with that of the existing method on complex eligibility criteria, 2) identifying opportunities and challenges for using GPT to automate query formulation, and 3) supporting human–computer interaction for interpreting generated SQL query and standardized OMOP concepts.

2. Methods

2.1. System overview

The overall workflow of C2Q 3.0 is presented in Fig. 1. OpenAI GPT-4 API was used in the system and the temperature was set to 0.0 for reproducibility [2]. We have implemented three one-shot GPT-prompts in the system: 1) concept extraction, 2) SQL generation, and 3) reasoning based on its demonstrated accuracy and consistency across a variety of use cases. The prompt architecture was adopted from previously proposed strategies [25]. All three prompts contain following content: task guidance, input specification, output specification, related documentation, and a few examples. The related documentation contains definitions of the OMOP domain, type of SQL languages, or database version. Fig. 2 and and Supplementary Table S1 show the prompt design, with the source code at https://github.com/Jimyung6642/Criteria2Query3.0.

We adopted the framework for cohort definition from OHDSI for user input collection, including the cohort entry event, inclusion, and exclusion criteria [19]. The inclusion and exclusion events can be free text or automatically extracted from the standardized trial protocol in ClinicalTrials.gov (https://clinicaltrials.gov). The cohort entry event or also called initial event, such as "patients diagnosed with hypertension", defines the starting cohort. The inclusion and exclusion criteria will be applied to the initial cohort. C2Q 3.0 recognizes clinical entities using GPT and leverages Usagi to map the entities to standardized concepts in OMOP-CDM [26]. Usagi is an automated concept-mapping software provided by Observational Health Data Sciences and Informatics (OHDSI). The user-editable interface was inherited from the previous versions of C2Q, allowing criteria selection, parsing error correction, and concept mapping. After concept mapping, the system formulates SQL queries and reasoning explanations behind the query and mapping. The outputs of the parsing, mapping, and reasoning processes are displayed with a user-editable interface. The interface allows editing the GPT-parsed clinical concepts, Usagi-mapped standardized concepts, and GPT-generated reasoning on formulated SQL queries.

The reasoning explanation can be edited and sent to GPT and used for re-formulating SQL queries. Our system uses Levenshtein distance to detect the alteration between original and the revised reasoning explanations. Modifications to reasoning explanations benefits two tasks. First, if the users want to incorporate additional criteria into their query (i.e., diagnosis, quantity of the medication, or demographics), they can directly add the criteria to the reasoning without modifying the original criteria. For example, "[Add criteria] Patients diagnosed with breast cancer" would yield the same results as adding this new criteria to the original criteria but is computationally more efficient because modifying the original criteria requires re-executing concept extraction and mapping modules. Second, if the users want to incorporate the logic between concepts, they can manually delineate the relationships between the concepts. For example, "[Add logic] Patients should have both hypertensive disorder AND type 2 diabetes mellitus. Once the SQL query is confirmed by the user, the system executes the query to the OMOP-CDM

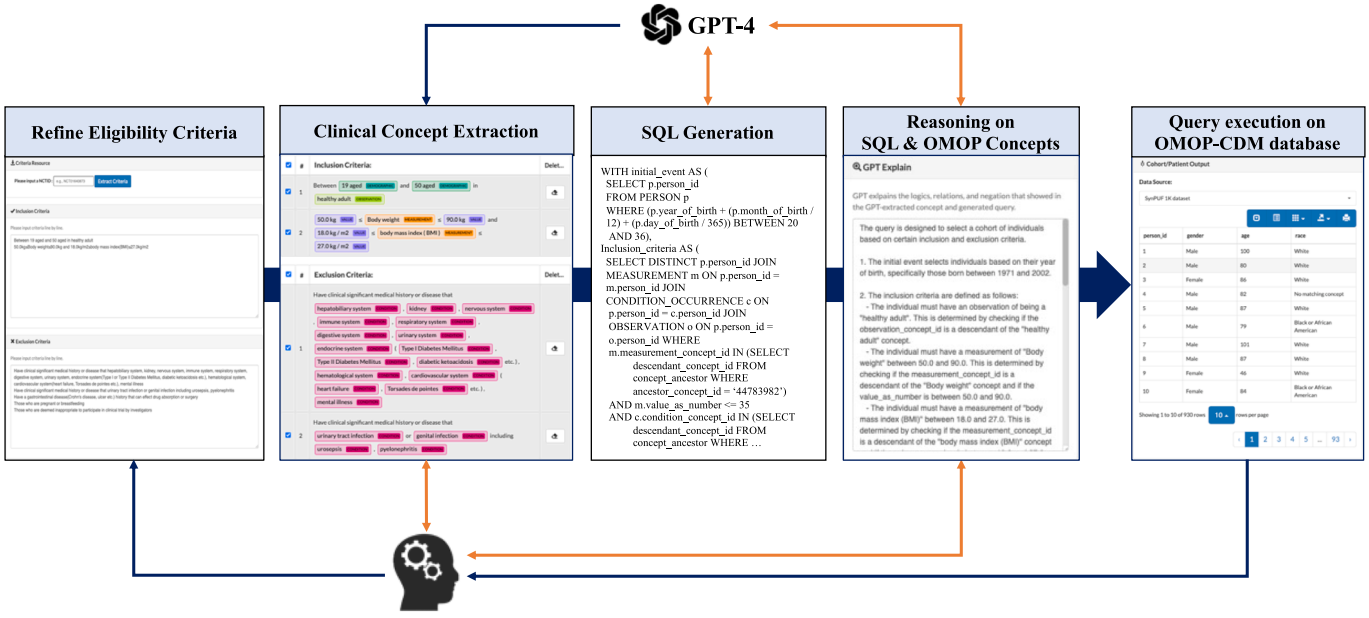


Fig. 1. Overall workflow of Criteria2Query 3.0 powered by GPT-4. The orange colored-arrows represent the interaction between human or GPT. If the users modify GPT-generated results, GPT will recognize the changes and re-execute each corresponding module. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

A) Concept Extraction

Task description	Annotate clinical concepts from the given text using the following rules: 1) Annotate concepts with …
Output format specification	Return your answer with extracted entities and under [Annotation] section.
Output example	Following is allowed examples: 1) <Measurement>eGFR</Measurement> <Value>triple negative</Value>
OMOP schema information	Condition is events of a Person suggesting the presence of a disease or medical condition stated …
Output	[initial_event] [inclusion_criteria] Patients diagnosed with <Condition>hypertension</Condition>

B) SQL Generation

Task description	Generate PostgreSQL query from the given text while following the rules: 1) Target database schema is OMOP …
Input format specification	The given text consists of three sections: [initial_event], [inclusion_criteria], and [exclusion cirteria] …
Output format specification	Use Common Table Expression and add semi-colon after each sql statement.
Output example	WITH initial_event AS (SELECT …your query…), inclusion_criteria AS (SELECT …your query…), …
Output	WITH initial_event AS (SELECT person_id from PERSON where gender_concept_id = 8532), inclusion_criteria

C) Reasoning

Task description	Explain logics, temporal, relation, and attributes between annotated concepts from a given query …
Input format specification	The query confirms to OMOP-CDM v5.3.2 and the concept mapping list consists of [data-entity], [concept-id] …
Output format specification	Return your answer as the narrative explanation with numbers. Explain the concepts with [concept-name] …
Output example	1. The initial_event is …your answer… 2. The inclusion_criteria is defined as …your answer…
Output	The given SQL query aims to identify a cohort of patients based on inclusion and exclusion criteria. The initial …

Fig. 2. Example architectures of three prompts in Criteria2Query 3.0.

formatted database and returns the cohort information.

2.2. Prompt design

2.2.1. Concept extraction

Fig. 2A shows the overall prompt design for concept extraction. In the concept extraction prompt, we described information extraction and normalization functionalities in the task description section. The functionalities include named entity recognition, negation detection, abbreviation normalization, and attribute normalization [18,20]. All entities were extracted and assigned to the categories in OMOP-CDM: Demographic, Condition, Drug, Device, Procedure, Measurement, and Observation. Additionally, we added Visit, Temporal, and Value categories that can be considered as the attributes of OMOP categories. To provide more flexibility in parsing complex criteria, we allowed GPT to create new categories for the recognized medical entities to better structure information. For example, the entities such as 'cycle of chemotherapy' and 'left/right site of kidney' cannot perfectly fit into any predefined categories above; GPT assigned 'drug cycle' and 'anatomic site', respectively. We also provided the definition of each category in OMOP-CDM tables, which is described in OMOP-CDM specifications [27] and the input and output data specifications in the prompt.

2.2.2. SQL generation

We implemented a text-to-SQL approach to generate SQL queries to retrieve the patient cohort conforming to the eligibility criteria. Fig. 2B presents the design of the SQL generation prompt. We adapted the existing prompt designs for text-to-SQL generation to our OMOP-specific SQL query generation [28,29]. For example, we instructed GPT to use standardized OMOP concept IDs while generating the query. The task description also includes considering the relations and logic lying in the eligibility criteria. To diminish syntax errors, we defined the output format to conform to the SQL dialect. As a one-shot prompting, an example query is provided. Information about what categories of entities can be found in each OMOP-CDM tables is also provided in the prompt to enhance the accuracy in identifying the appropriate tables for search records of clinical concepts. For example, Demographic information can be extracted from the PERSON table and Condition can be extracted from the CONDITION_OCCURRENCE table.

2.2.3. Reasoning

The target users of C2Q 3.0 can be clinical research coordinators or clinical research staff members. Our system does not directly present SQL queries to the users. Only the criteria parsing result and its reasoning and the list of eligible patients with their demographics (i.e., gender, age, and race) are presented in the system. This prompt aims to improve the system's interpretability by explaining to the users how the system generates queries from the given eligibility criteria. Specifically, the prompt explains the relations between concepts, the logic between the eligibility criteria, and values or units of lab measurements, which cannot be identifiable in the current user interface. Since we adopted the GPT-based SQL generation process, the generated SQL query could have errors and biases, requiring a human-in-the-loop process to minimize the selection bias while pre-screening patients. Particularly for individuals lacking knowledge in data structures, SQL, or the OMOP-CDM, the human-like language generation of GPT can facilitate understanding of the system processing on parsing and queries [30]. Similar to the other

prompts, we adapted the preceding prompting strategy, incorporating OMOP-specific knowledge [31]. The task was set to narratively describe the logic, relation, negation, and other attributes of the observed concepts in the query (Fig. 2C). In this prompt, GPT-generated SQL query and extracted entities with standardized OMOP concept IDs are provided as input data. The output was a narrative explanation with numbered paragraphs to enhance readability.

2.3. Evaluation

The tasks were evaluated independently with clinical trials collected from ClinicalTrials.gov. We purposely selected twenty clinical trials covering a variety of conditions, interventions, and trial phases to perform a comprehensive and rational evaluation. Trials with improper format style, vague clinical abbreviation, or which do not include both inclusion and exclusion criteria were excluded. For example, trials that do not distinguish inclusion or exclusion criteria are excluded, and trials including abbreviations without definitions such as FC (potentially meaning fevers and/or chills, functional class, etc.) or HSS (potentially meaning hyperosmolar hyperglycemic state or hepatic stimulator substance) are excluded. To measure the evaluation adequacy of the trials, we calculated the complexity of the eligibility criteria, which measures the number of complex patterns of clinical concept in a single phrase and trimmed the trials with complexity scores less than 0.1 (extremely simple) or above 0.9 (extremely complex) [32]. Overall, twenty clinical trials of 18 diseases were selected and the average complexity of the selected trials was identified as 0.46 (Supplementary Table S2). Five clinical trials whose free-text eligibility criteria were successfully translated to OMOP-CDM (NCT05880693 NCT05719155, NCT05521633, NCT05535595, and NCT05479214) were used for the evaluation of the SQL query generation and GPT-reasoning. Eligibility criteria with missing information or vague index date were excluded from the evaluation trials since they can lead to misinterpretation. For example, “FVC less than 40 % of predicted at baseline” or “2 weeks prior to the start of screening period” do not contain baseline information, which can lead to incorrect queries. Due to the extensive effort required for these tasks, only five trials were evaluated here.

Overall, seven evaluators participated in the evaluation. Two biomedical informaticians evaluated the concept extraction and SQL generation tasks were each evaluated by two biomedical informaticians, and three clinicians evaluated the reasoning task. For concept extraction, reviewers were instructed to annotate clinical concepts with OMOP categories, and their consensus results were considered the gold standard and were compared with the system-extracted concepts. Cohen’s kappa coefficient was calculated to assess the inter-rater agreement. In addition, we compared the concept extraction performance of the current system (GPT-4) with GPT-3.5-turbo and with C2Q 2.0. Precision, recall, and F1-score are used as evaluation metrics. For SQL generation, evaluators first manually reviewed the GPT-generated queries to identify semantic errors (e.g., criteria misinterpretation, function misuse, or concept omissions) and executed them on the OMOP-CDM database to confirm the validity of the queries. To evaluate GPT-generated reasoning, we developed a 17-item questionnaire consisting of four categories: readability, correctness, coherence, and usefulness. The evaluators were instructed to compare the target free-text eligibility criteria and GPT-generated reasoning and to score the questions. The items were rated using a 5-point Likert scale, ranging from “Strongly disagree” as (1) to “Strongly agree” as (5).

3. Results

3.1. Concept extraction

Overall, 518 entities were identified from twenty clinical trials by two evaluators (Supplementary Figure S1). In our study, the inter-rater agreement measured by the kappa coefficient achieved a substantial agreement of 0.75 [33]. The most frequent entity category was Condition, with a total of 206 (39.77 %) occurrences. Value and Observation were the following most frequent categories, accounting for 78 (15.06 %) and 59 (11.39 %) occurrences, respectively. On the other hand, Device (1; 0.19 %) and Visit (2; 0.39 %) were identified as the least frequent categories in all criteria.

Table 1 presents a comparative analysis of the micro level performance among the three systems: C2Q 3.0 with GPT-4, C2Q 3.0 with GPT-3.5-turbo, and C2Q 2.0. C2Q 3.0 with GPT-4 resulted in the highest scores across all evaluation metrics, with a precision of 0.863, recall of 0.922, and F1-score of 0.891. Meanwhile, C2Q 3.0 with GPT-3.5-turbo achieved the lowest recall (0.534) and F1-score (0.612). The macro level performance by OMOP categories is described in Supplementary Table S3. Categories with relatively high frequency in the eligibility criteria (i.e., Condition, Drug, Measurement, Value, Temporal, and Procedure) achieved high performance (F1-score > 0.800). On the other hand, categories with lower frequency (i.e., Device and Visit) achieved low performance (F1-score ≤ 0.500). The macro level F1-score of C2Q 3.0 with GPT-4 was identified as 0.764 and weighted F1-score was identified as 0.910.

3.2. SQL generation

Overall, five SQL queries corresponding to the five evaluation trials were generated from the system. Among the five generated queries, we identified 29 overall errors, with seven different error types. The specific error types and their definitions are presented in Table 2. We classified three types of semantic error as logic error, concept omission, and incorrect concept mapping. Logic error was defined as the error related to the misinterpretation of the relation, temporal, quantity, and negation expression in the criteria. For example, misinterpreting age criteria or switching the values between two different lab measurements were classified as logic errors. Concept-related errors were defined as missing concepts that identified concepts in the concept extraction process but were missed in SQL formulation. Another concept-related error was the incorrect concept mapping, which resulted from Usagi, the adopted framework for concept mapping. The other errors (i.e., functional misuse, integrity constraint violation, schema reference error, and syntax error) were structural errors related to the foundational structure of the SQL query. The distribution of the errors across the five clinical trials is presented in Fig. 3. Logic error was identified as the most prevalent error with 10 (34.48 %) occurrences, followed by concept missing and incorrect concept mapping errors, each with 6 (20.69 %) occurrences, respectively. The most common logic error was identified as duplicated patient records in the cohort ($n = 5$), resulting from missing the DISTINCT statement in the query. In contrast, integrity constraint violation and schema reference error were identified as the least frequent errors, both with 1 (3.45 %) occurrence. Meanwhile, the clinical trial (NCT05521633), whose eligibility criteria achieved a relatively high complexity score (0.64), had the most errors ($n = 10$). Conversely, the trial (NCT05719155), whose criteria were with a low complexity score (0.29), had the fewest errors ($n = 2$).

3.3. Reasoning

Table 3 and Supplementary Figure S2 present the evaluation of the reasoning task based on four main categories: readability, correctness,

Table 1
Performance comparison on concept extraction among different versions of the Criteria2Query system.

System	Precision	Recall	F1-score
Criteria2Query 3.0 with GPT-4	**0.863**	**0.922**	**0.891**
Criteria2Query 3.0 with GPT-3.5	0.717	0.534	0.612
Criteria2Query 2.0	0.667	0.753	0.707

Table 2
Identified error types and definitions from GPT-generated SQL query.

Error type	Definition
Concept missing	Concept missing error means a clinical concept is detected at the concept extraction stage but not used in query formulation.
Functional misuse error	Functional misuse error means that SQL functions are incorrectly used in the query and eventually cause the error.
Incorrect concept mapping	Incorrect concept mapping means that extracted clinical concepts are mapped to the wrong standardized OMOP vocabulary.
Integrity constraint violation	Integrity constraint violation means that the query violates the OMOP's data consistency, such as inserting string value into the columns expecting integer concept IDs.
Logic error	Logic error means that GPT incorrectly defines the logic, relation, or temporality of the clinical concepts in the query.
Schema reference error	Schema reference error means that GPT incorrectly referenced OMOP-CDM tables or columns.
Syntax error	Syntax error is an error that has incorrect SQL, syntax such as misspelled words or incorrect use of punctuation in the query.

coherence, and usefulness. The mean score of readability questions was 3.95 (standard deviation [SD] = 0.64). The clarity of the initial events, inclusion, and exclusion criteria was rated, ranging from 3.22 to 3.53, which suggests that GPT's understanding of initial event, inclusion, and exclusion criteria can be relatively accurate. The clarity of layout, writing format, concepts, and terminologies were rated, ranging from 4.33 to 4.80. The mean score of correctness was 3.97 (SD = 0.58). The appropriate use of clinical terminologies and concepts was scored as 3.60 (SD = 0.80), and the completeness of the concepts was scored as 3.33 (SD = 0.90), meaning some of the clinical concepts in the protocol are not mentioned in GPT-generated reasoning which is caused by the concept missing error in SQL query. Meanwhile, the logic, relation, and attribute between concepts were rated, ranging from 4.40 to 4.60. The mean score of coherence was 4.70 (SD = 0.14). Coherence was scored highly, particularly on the clear explanation of the relations between the different criteria of the trial. Lastly, the mean score of usefulness was 4.37 (SD = 0.52), indicating that the reasoning prompt aids in understanding the system and the clinical trial, as well as being a useful resource for modifying or reproducing the query. The only aspect scoring lower in usefulness was the presentation of the OMOP concepts, rated as 3.67 (SD = 0.95).

4. Discussion

C2Q 3.0 leverages OpenAI GPT-4 to enhance accurate criteria parsing and query formulation from free-text eligibility criteria and uses GPT-generated reasoning prompt to explain SQL queries and standardized concepts. It automates the parsing of free-text eligibility criteria and the generation of SQL cohort queries for standardized clinical databases. GPT-4 outperformed the C2Q 2.0 and GPT-3.5-turbo in criteria parsing.

The most prevalent errors of GPT-generated queries were logic and concept-related errors, and complex criteria tend to generate more errors. In concept extraction, GPT-4 showed the highest performance in all evaluation metrics. Most of the false positives that lower precision, e.g., "*English speaking*", "*patient's willingness*", or "*patient's agreement*", are clinical entities that are not queryable in OMOP databases. These concepts are easily comprehensible for humans only [34,35]. Further studies can benefit from efficient classification of queryable and non-queryable clinical concepts in criteria.

Meanwhile, GPT-3.5-turbo showed lower performance and even failed to conform to the output format instruction in the prompt, consistent with the results from other studies (Supplementary Figure S3) [15,25,36–38]. The studies also suggested that traditional language models fine-tuned on a specific domain can achieve higher performance than GPT [25,39,40]. However, comparing fine-tuned language models with few-shot or zero-shot learned GPT might be unfair. OpenAI recently released fine-tuning APIs for GPT-3.5 and GPT-4, and system performance can be improved in the future through fine-tuning. Other studies also suggested that GPT-4 without fine-tuning outperformed the C2Q 2.0 which leverages traditional language models trained on clinical trial eligibility criteria corpus [18,20]. The performance of LLMs in biomedical research is still controversial and requires further evaluation in several aspects [41]. We identified that C2Q 3.0 with GPT-4 can accurately recognize concepts in complex criteria, compared to GPT-3.5-turbo or C2Q 2.0 (Supplementary Figure S3). For trial NCT05719155, while C2Q 2.0 could not distinguish between serially mentioned measurements and their corresponding values, GPT-4 correctly identified each Measurement entity. GPT-3.5-turbo could recognize the exact spans of the entities, but it failed to annotate the correct OMOP category. In NCT05532800, C2Q 2.0 was unable to identify CABG, while GPT-4 was able to identify all entities and was able to perform abbreviation normalization. However, the system using GPT-3.5-turbo was not adhering to the specified output format and failed to recognize certain entities with a correct OMOP category. GPT-4 also correctly identified

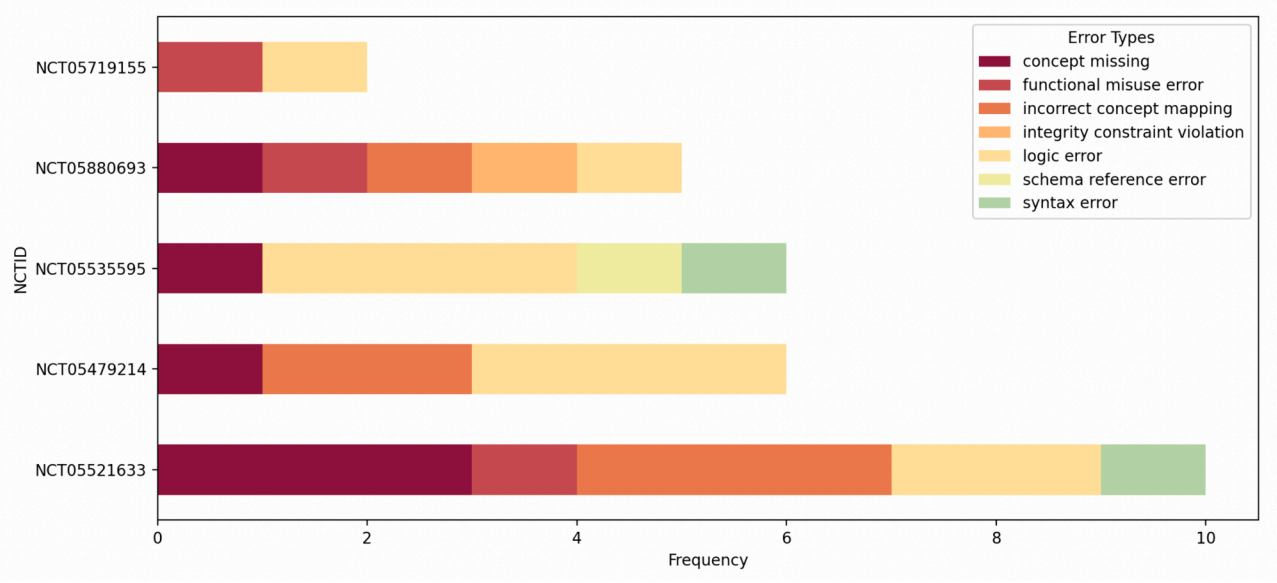


Fig. 3. Manually identified SQL errors from GPT-generated queries on five clinical trials.

Table 3
Questionnaire and mean scores on GPT-generated reasoning explanation.

Category	Question	Mean score (standard deviation)
Readability	Overall writing-format is easy to understand.	4.80 (0.35)
	The layout of the answer is easy to follow, such as numbered points and bullet points.	4.40 (0.53)
	The clinical concepts and terminologies are easy to understand.	4.33 (0.50)
	The answer clearly explains the inclusion criteria of the clinical trial	3.53 (0.70)
	The answer clearly explains the exclusion criteria of the clinical trial	3.40 (0.69)
	The answer clearly explains the initial event, if exists, of the clinical trial	3.22 (1.02)
Correctness	The logics between concepts are clearly explained.	4.60 (0.35)
	The relations between concepts are clearly explained	4.53 (0.12)
	The attributes (e.g., values of measurement) between concepts are clearly explained	4.40 (0.40)
	The answer accurately uses the appropriate clinical terminologies and concepts.	3.60 (0.80)
	All clinical concepts mentioned in the trial protocol are mentioned in the explanation.	3.33 (0.90)
Coherence	The relations between initial, inclusion, and exclusion criteria are correctly explained.	4.80 (0.20)
	The answer consistently maintains the same level of detail when explaining the different parts of the query.	4.60 (0.53)
Usefulness	Explaining the logic, relation, and attributes between concepts are helpful to understand which patients are queried.	4.80 (0.35)
	The answer is helpful to understand how the system understand the clinical trial and generated the query.	4.73 (0.31)
	The answer is useful if you want to modify or reproduce the query.	4.27 (0.70)
	Presenting the OMOP concepts used in the query is helpful to understand which patients are queried.	3.67 (0.95)

the clinical abbreviation as follows: "MI" as "Myocardial Infarction," "UA" as "Unstable Angina," "PCI" as "Percutaneous Coronary Intervention," and "CABG" as "Coronary Artery Bypass Graft." We recognize the potential value in evaluating negation detection and abbreviation normalization and have included them as future work. Supplementary Figure S4 also presents that C2Q 3.0 can accurately identify the measurement-related concepts and values and multiple concepts of different categories in a single sentence.

The errors identified from GPT-generated SQL queries were mostly related to logic and concepts, which are critical for accurate patient retrieval. Since OpenAI maintains GPT API, we could not identify the cause of the errors in this study. One possible reason for the errors is the performance variation given the length of the prompt context [42,43]. As the SQL generation prompt already contains 1,188 tokens, GPT might not be able to capture all concepts and the logic properly, resulting in the omission of the concepts in the query. Another reason might be the performance changes of the OpenAI LLMs over time. Chen et al. assessed the performance of GPT models from March to June 2023 and observed significant performance variation [17]. Considering that the development period of the system overlaps with their study period (from March to June 2023), the performance variation over time might influence our study result. Despite the current limitations, generative LLM's text-to-SQL performance is still impressive and can be beneficial [28]. The most frequent logic error was the missingness of the DISTINCT statement, which can be considered relatively not critical for cohort retrieval and can be revised with clear instructions in the criteria. Incorrect concept mapping is critical, but it was caused by the Usagi, which is out of the scope of our study. However, other logic-related errors, such as missing lab values or not specifying AND/OR in the SQL query, might be critical and, therefore, require further investigation in future studies. One possible solution to address the issue is simplifying the eligibility criteria before they are processed by C2Q 3.0. We implemented a built-in interactive user interface to allow users to simplify the eligibility criteria. Users can edit the criteria before processing them and uncheck the parsed criteria based on their decision. Additionally, we investigated potential performance improvement through the criteria rephrasing (Supplementary Figure S5) [44]. The results showed that rephrasing the eligibility criteria can improve concept extraction performance, minimizing SQL errors.

To efficiently automate the simplification, identifying complexity of the eligibility criteria is required. Ross et al. has identified the complex patterns of the eligibility criteria [32]. Based on the patterns, complex criteria can be automatically identified and simplified through the rephrasing module, which can enhance overall performance of C2Q 3.0. Future work could consider implementing the module focusing on the criteria simplification. Another solution can be adopting a self-correction approach, which may reduce errors related to the validity of SQL queries [45,46]. Although we did not systematically assess GPT's ability to self-correct SQL queries, LLM's self-correction has been shown to improve query accuracy in previous studies [29,47]. Besides, except for syntax-related errors, the overall structures of GPT-generated SQL queries were concise and could be executed without human correction. Considering C2Q interface integrates a human-in-the-loop process, GPT-based semi-automated query generation, correction, and execution can be beneficial for gradually enhancing accuracy. Furthermore, rule-based query formulation which was used in C2Q 2.0 can be inherited in the future version allowing complementary with GPT-generated query. For example, we can further use a rule-based formulated query to fine-tune GPT or set it as an example query in the prompt.

Through the reasoning evaluation, we confirmed that GPT can help users understand GPT-generated queries and standardized concepts. Considering that some C2Q users are not familiar with SQL or OMOP CDM and concept normalization, this narrative explanation can assist them in understanding how the system processes the given eligibility criteria [21–23]. However, the readability and correctness of the explanations achieved lower scores. Particularly, concept omissions, presentation of concept IDs, or inaccurate concept names hindered comprehension of the explanations. Nevertheless, the reasoning explanation gives users a narrative description of how the system identifies the complex relations between the parsed clinical concepts and structured SQL queries to extract the cohort to prevent biased cohort extraction. From our real-world experience adopting C2Q 2.0, we received the feedback from the end-users who worried about the lack of validation process of the generated query. Even though the generated query can be downloaded from the system, it is only helpful if they can understand the SQL query. Therefore, we implemented a reasoning explanation in C2Q 3.0, responding to the user's requirement. As GPT can generate human-like text, users can clearly understand how the system defines the relation and logic between concepts and extracts data. Future studies should investigate the interaction between human users and the outputs generated by systems using LLM.

This study has several limitations. First, as we are using OpenAI's API, we were unable to address certain issues such as model performance inconsistency and latency. Moreover, we observed that GPT occasionally did not follow prompt instructions and generated results in a random format. Leveraging open-source LLMs in a local environment might address the issues and be more beneficial in ensuring system maintenance and data security. Previously, we could not identify the optimal open-source LLM for C2Q 3.0, which showed robust performance over three tasks. Fine-tuning LLMs may increase the performance; however, fine-tuning the models for each task will require extensive computing resources. Therefore, we concluded that we should leverage GPT and leave the task for future work. In the future, we will investigate the optimal open-source LLMs to deploy in a local

environment to prevent performance variation over time [48]. Second, C2Q 3.0 requires improvements in user experience. The reasoning interface was implemented to improve the interpretability of the system, but offering a narrative description could potentially lead to information overload for users [49]. Furthermore, users can edit the criteria parsing results based on their preferences; however, this can also increase the user's burden, especially regarding the removal of unnecessary criteria. Third, the number of clinical trials for evaluation is limited. Subsequent research should expand its scope by assessing more clinical trials with greater diversity. Fourth, the mapping module needs to be enhanced for the practical use of C2Q 3.0. Even with outstanding concept extraction performance, the incorrect mapping to the standardized vocabulary can eventually cause errors in the query. For example, our system can successfully extract the diagnostic modifiers; however, the mapping module only correctly maps 6 out of 12 extracted concepts (Supplementary Figure S6). Overall, the current LLM-based system for generating queries for trial cohort identification is an experimental approach that requires more evaluation to prove its usability and robustness in a real setting. In the future, more comprehensive evaluation data and metrics will be required for each task.

5. Conclusion

This study demonstrated that C2Q 3.0 powered by OpenAI GPT-4 can significantly enhance the accuracy for concept extraction, query formulation, and reasoning of the eligibility criteria. Our approach, consisting of three GPT modules, can identify patient cohorts meeting the eligibility criteria while keeping patient records in the local environment without transferring OpenAI. Leveraging LLM might be challenging, particularly regarding SQL generation and reasoning. Nevertheless, it has already shown a promising future, surpassing what traditional methods can achieve. Further investigation is required to confirm the system's stable and consistent use of generative LLM.

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Problem or issue

Free text clinical trial eligibility criteria are not amenable for computer processing. There is a need for scalable, explainable, and accurate method for automatically transforming criteria text into standards-based cohort database queries.

What is already known

Generative large language models can be beneficial for several biomedical natural language processing tasks including concept identification from the eligibility criteria.

What this paper adds

Criteria2Query 3.0 leveraged GPT-4 to improve overall processing for semi-automatic parsing of the eligibility criteria of the clinical trials.

CRediT authorship contribution statement

Jimyung Park: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Yilu Fang:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Casey Ta:** Writing – review & editing, Validation, Methodology, Investigation. **Gongbo Zhang:** Writing – review & editing, Validation, Conceptualization. **Betina Idnay:** Writing – review & editing, Validation. **Fangyi Chen:** Writing – review & editing, Validation. **David Feng:** Writing – review & editing, Validation. **Rebecca Shyu:** Writing – review & editing, Validation. **Emily R. Gordon:** Writing – review & editing, Validation. **Matthew Spotnitz:** Writing – review & editing, Validation. **Chunhua Weng:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jbi.2024.104649.

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